MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 Very Short Answer (VSA) question of 2 marks.
4. **Section C** comprises 3 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 1 Applied Case-Based question of 4 marks with internal choice.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. From each corner of a square of side 4 cm, a quadrant of a circle of radius 1 cm is cut. The sum of the areas of the four quadrants is: (Use $\pi = 3.14$) [1]

(A) 3.14 cm^2 (B) 6.28 cm^2 (C) 12.56 cm^2 (D) 1.57 cm^2

Q2. Four equal circles of radius a touch each other externally. The area enclosed between the four circles is: [1]

(A) $(4 - \pi)a^2$ (B) $(2 - \pi)a^2$ (C) $(\pi - 2)a^2$ (D) $(4 - 2\pi)a^2$

Q3. A racetrack is in the form of a ring. If the difference between the outer and inner circumferences is 44 m, then the width of the track is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 14 m (B) 7 m (C) 3.5 m (D) 22 m

Q4. If the radius of a circle is increased by 50%, then its area is increased by: [1]

(A) 50% (B) 100% (C) 125% (D) 150%

Q5. **Assertion (A):** If a circle circumscribes a square of side a, then the area of the circle is $\frac{\pi a^2}{2}$.
Reason (R): The diagonal of the inscribed square coincides with the diameter of the circumscribing circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (1 × 2 = 2 Marks)

Q6. Find the area of the sector of a circle of radius 7 cm if the length of the corresponding arc is 11 cm. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (3 × 3 = 9 Marks)

Q7. A car has two windshield wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115°. Calculate the total area cleaned at each sweep of the blades. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

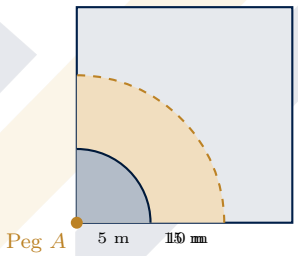
Q8. A calf is tied with a rope of length 6 m at one corner of a rectangular grassy lawn of dimensions 20 m × 16 m. Find: (i) the area of the lawn the calf can graze, and (ii) the increase in grazing area if the rope length is increased to 8 m. (Use $\pi = 3.14$) [3]

Q9. An archery target has its gold region with diameter 21 cm and each of the outer concentric bands is 10.5 cm wide. Find: (i) the area of the gold region, and (ii) the area of the red band. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

SECTION D: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q10. Pasture Grazing Dynamics on a Square Meadow

A horse is tethered with a rope to a corner peg of a square fenced grass meadow of side 15 m. Initially, the rope length is 5 m. Later, the farmer extends the rope length to 10 m to permit access to more grazing area.



(i) What geometric shape is swept by the horse while grazing within the meadow boundary? [1]

(ii) Find the initial grazing area accessible to the horse with the 5 m rope. [1]

(iii) Calculate the increase in the grazing area when the rope is extended to 10 m, OR find the remaining ungrazed area of the meadow when the rope is 10 m long. (Use $\pi = 3.14$) [2]

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